

Real World Connection 1.5

Objective: My objective in this Virtual lab was to pick up on my “Real world connection 1” project and showcase the skills that I wasn't able to show off due to the limitations that Packet Tracer presents. Instead of doing a 3-core layer design, I decided to make a collapsed core design for this topology. Each building has its own pair of VLANs that contain just end hosts. Since this is a collapsed core topology I wanted the routers to have full control of the network, so I made ROAS configuration to stay true to that. I excluded the last available IP addresses of every VLAN to have a static IP for the default gateway and made the router the DHCP Servers for their respective VLAN. As for the layer 2 switches, I configured every security feature possible within the network devices. I made sure to include ARP inspection, DHCP snooping and Port-Security, with Building A and C I configured PAT on the router with the interfaces that are facing outwards. I included ACL rules to block Building A and C from communicating. Lastly, I used OSPF to allow full communication.

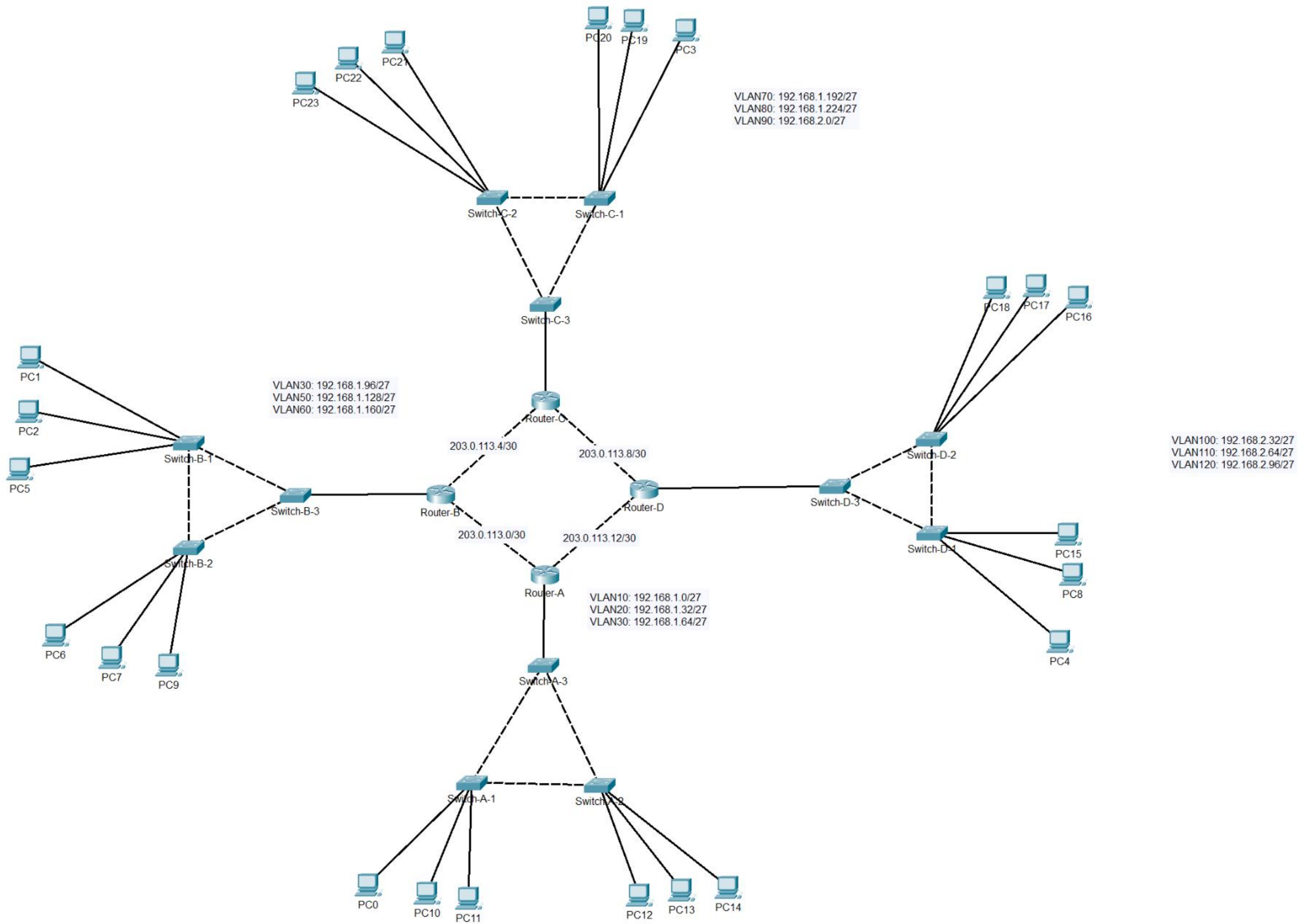
Equipment: (4) 2911 Routers, (12) 2960 Switches, and (24) PC's

Key Steps:

- a. Divide the network 192.168.1.0 using VLSM to make sure every subnet has an equal host of 30
- b. Make a set of 3 VLANs per network
- c. Create trunk links between the switches and routers
- d. Create a DHCP Pool within each VLAN, excluding the last address for the default gateway
- e. Apply PAT configurations on router A and C
- f. Enable ARP Inspection on all interconnected network devices, made sure to validate dst-mac, src-mac and ip
- g. Enable DHCP Snooping on each VLAN, while trusting every interface that is going upstream the network
- h. Enable Port-Security on every access port within every switch with a maximum of 1 MAC address
- i. Configured OSPF to allow full network communication

Drawbacks :

The drawbacks from this network design is that there are no channel-groups present to allow for faster speeds, this was due to Packet Tracer limitations that didn't allow for etherchannels and ARP Inspection with DHCP Snooping to be combined. Another flaw is that since this is a collapsed core topology, routers are in charge of everything like ROAS, ACL, and DHCP server which deteriorates the network speed as well. I also did want PAT to work within every internal network but Packet Tracer limitations prevented 2 networks that were configured properly from communicating with each other, so I implemented ACLs to simulate a real world scenario. My main goal was just to touch on the skills that I couldn't touch on like DHCP Snooping, ARP Inspection, ROAS, and PAT.



Real World Connection 1.5 (interfaces)

Switch-A-1	F0/1	Trunk Port	Switch-A-2
Switch-A-1	F0/2	Trunk Port	Switch-A-3
Switch-A-1	F0/3	PC 0	DHCP
Switch-A-1	F0/4	PC 10	DHCP
Switch-A-1	F0/5	PC 11	DHCP
Switch-A-2	F0/1	Trunk Port	Switch-A-1
Switch-A-2	F0/2	Trunk Port	Switch-A-3
Switch-A-2	F0/3	PC 12	DHCP
Switch-A-2	F0/4	PC 13	DHCP
Switch-A-2	F0/5	PC 14	DHCP
Switch-A-3	F0/1	Trunk Port	Switch-A-1
Switch-A-3	F0/2	Trunk Port	Switch-A-2
Switch-A-3	F0/3	Trunk Port	Router-A
Router-A	G0/0.10	ROAS	192.168.1.30
Router-A	G0/0.20	ROAS	192.168.1.62
Router-A	G0/0.30	ROAS	192.168.1.94
Router-A	G0/1	Router-B	203.0.113.1
Router-A	G0/2	Router-D	203.0.113.14
Switch-B-1	F0/1	Trunk Port	Switch-B-2
Switch-B-1	F0/2	Trunk Port	Switch-B-3
Switch-B-1	F0/3	PC 1	DHCP
Switch-B-1	F0/4	PC 2	DHCP
Switch-B-1	F0/5	PC 5	DHCP
Switch-B-2	F0/1	Trunk Port	Switch-B-1
Switch-B-2	F0/2	Trunk Port	Switch-B-3
Switch-B-2	F0/3	PC 6	DHCP
Switch-B-2	F0/4	PC 7	DHCP
Switch-B-2	F0/5	PC 9	DHCP
Switch-B-3	F0/1	Trunk Port	Switch-B-1
Switch-B-3	F0/2	Trunk Port	Switch-B-2
Switch-B-3	F0/3	Trunk Port	Router-B
Router-B	G0/0.40	ROAS	192.168.1.126
Router-B	G0/0.50	ROAS	192.168.1.158
Router-B	G0/0.60	ROAS	192.168.1.190
Router-B	G0/1	Router-A	203.0.113.2
Router-B	G0/2	Router-C	203.0.113.5

Switch-C-1	F0/1	Trunk Port	Switch-C-2
Switch-C-1	F0/2	Trunk Port	Switch-C-3
Switch-C-1	F0/3	PC 3	DHCP
Switch-C-1	F0/4	PC 19	DHCP
Switch-C-1	F0/5	PC 20	DHCP
Switch-C-2	F0/1	Trunk Port	Switch-C-1
Switch-C-2	F0/2	Trunk Port	Switch-C-3
Switch-C-2	F0/3	PC 21	DHCP
Switch-C-2	F0/4	PC 22	DHCP
Switch-C-2	F0/5	PC 23	DHCP
Switch-C-3	F0/1	Trunk Port	Switch-C-1
Switch-C-3	F0/2	Trunk Port	Switch-C-2
Switch-C-3	F0/3	Trunk Port	Router-C
Router-C	G0/0.70	ROAS	192.168.1.222
Router-C	G0/0.80	ROAS	192.168.1.254
Router-C	G0/0.90	ROAS	192.168.2.30
Router-C	G0/1	Router-B	203.0.113.6
Router-C	G0/2	Router-D	203.0.113.9
Switch-D-1	F0/1	Trunk Port	Switch-D-2
Switch-D-1	F0/2	Trunk Port	Switch-D-3
Switch-D-1	F0/3	PC 4	DHCP
Switch-D-1	F0/4	PC 8	DHCP
Switch-D-1	F0/5	PC 15	DHCP
Switch-D-2	F0/1	Trunk Port	Switch-D-1
Switch-D-2	F0/2	Trunk Port	Switch-D-3
Switch-D-2	F0/3	PC 16	DHCP
Switch-D-2	F0/4	PC 17	DHCP
Switch-D-2	F0/5	PC 18	DHCP
Switch-D-3	F0/1	Trunk Port	Switch-D-1
Switch-D-3	F0/2	Trunk Port	Switch-D-2
Switch-D-3	F0/3	Trunk Port	Router-D
Router-D	G0/0.100	ROAS	192.168.2.62
Router-D	G0/0.110	ROAS	192.168.2.94
Router-D	G0/0.120	ROAS	192.168.2.126
Router-D	G0/1	Router-C	203.0.113.10
Router-D	G0/2	Router-D	203.0.113.13

Switch-A-1:

```
!  
hostname Switch-A-1  
!  
ip arp inspection vlan 10,20,30  
ip arp inspection validate src-mac dst-mac ip  
!  
ip dhcp snooping vlan 10,20,30  
no ip dhcp snooping information option  
ip dhcp snooping  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
spanning-tree vlan 20 priority 24576  
!  
interface FastEthernet0/1  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 10,20,30  
  ip arp inspection trust  
  switchport mode trunk  
!  
interface FastEthernet0/2  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 10,20,30  
  ip arp inspection trust  
  ip dhcp snooping trust  
  switchport mode trunk  
!  
interface FastEthernet0/3  
  switchport access vlan 10  
  switchport mode access  
  switchport port-security  
  switchport port-security aging time 60  
!  
interface FastEthernet0/4  
  switchport access vlan 20  
  switchport mode access  
  switchport port-security  
  switchport port-security aging time 60  
!  
interface FastEthernet0/5  
  switchport access vlan 30  
  switchport mode access  
  switchport port-security  
  switchport port-security aging time 60  
!  
interface FastEthernet0/6  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/7  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/8  
  switchport access vlan 30  
  switchport mode access  
!  
interface FastEthernet0/9
```

```

switchport access vlan 10
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 30
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
end

```

Switch-A-2:

```

!
hostname Switch-A-2
!
ip arp inspection vlan 10,20,30
ip arp inspection validate src-mac dst-mac ip
!
ip dhcp snooping vlan 10,20,30
no ip dhcp snooping information option
ip dhcp snooping
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 30 priority 24576
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30
ip arp inspection trust
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk
!
interface FastEthernet0/3
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security aging time 60
!

```

```

interface FastEthernet0/4
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/5
switchport access vlan 30
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/6
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/7
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/8
switchport access vlan 30
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 30
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
end

```

Switch-A-3:

```

!
hostname Switch-A-3
!
ip arp inspection vlan 10,20,30
ip arp inspection validate src-mac dst-mac ip
!
ip dhcp snooping vlan 10,20,30
no ip dhcp snooping information option

```

```

ip dhcp snooping
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 20,30 priority 20480
spanning-tree vlan 10 priority 24576
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30
ip arp inspection trust
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30
ip arp inspection trust
switchport mode trunk
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
end

```

Switch-B-1:

```

!
hostname Switch-B-1
!
ip arp inspection vlan 40,50,60
ip arp inspection validate src-mac dst-mac ip
!
ip dhcp snooping vlan 40,50,60
no ip dhcp snooping information option
ip dhcp snooping
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 40 priority 24576
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60
ip arp inspection trust

```

```
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk
!
interface FastEthernet0/3
switchport access vlan 40
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/4
switchport access vlan 50
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/5
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/6
switchport access vlan 40
switchport mode access
!
interface FastEthernet0/7
switchport access vlan 50
switchport mode access
!
interface FastEthernet0/8
switchport access vlan 60
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 40
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 50
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 60
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
```

```
!  
end
```

Switch-B-2:

```
!  
hostname Switch-B-2  
ip arp inspection vlan 40,50,60  
ip arp inspection validate src-mac dst-mac ip  
!  
ip dhcp snooping vlan 40,50,60  
no ip dhcp snooping information option  
ip dhcp snooping  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
spanning-tree vlan 50 priority 24576  
!  
interface FastEthernet0/1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60  
ip arp inspection trust  
switchport mode trunk  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60  
ip arp inspection trust  
ip dhcp snooping trust  
switchport mode trunk  
!  
interface FastEthernet0/3  
switchport access vlan 40  
switchport mode access  
switchport port-security  
switchport port-security aging time 60  
!  
interface FastEthernet0/4  
switchport access vlan 50  
switchport mode access  
switchport port-security  
switchport port-security aging time 60  
!  
interface FastEthernet0/5  
switchport access vlan 60  
switchport mode access  
switchport port-security  
switchport port-security aging time 60  
!  
interface FastEthernet0/6  
switchport access vlan 40  
switchport mode access  
!  
interface FastEthernet0/7  
switchport access vlan 50  
switchport mode access  
!
```

```

interface FastEthernet0/8
switchport access vlan 60
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 40
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 50
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 60
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
end

```

Switch-B-3:

```

!
hostname Switch-B-3
!
ip arp inspection vlan 40,50,60
ip arp inspection validate src-mac dst-mac ip
!
ip dhcp snooping vlan 40,50,60
no ip dhcp snooping information option
ip dhcp snooping
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 40,50 priority 20480
spanning-tree vlan 60 priority 24576
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60
ip arp inspection trust
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60
ip arp inspection trust
switchport mode trunk
!
interface FastEthernet0/3

```

```
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk
```

```
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
end
```

Switch-C-1:

```
!
hostname Switch-C-1
!
ip arp inspection vlan 70,80,90
ip arp inspection validate src-mac dst-mac ip
!
ip dhcp snooping vlan 70,80,90
no ip dhcp snooping information option
ip dhcp snooping
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 70 priority 24576
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90
ip arp inspection trust
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk
!
interface FastEthernet0/3
switchport access vlan 70
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/4
switchport access vlan 80
switchport mode access
```

```

switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/5
switchport access vlan 90
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/6
switchport access vlan 70
switchport mode access
!
interface FastEthernet0/7
switchport access vlan 80
switchport mode access
!
interface FastEthernet0/8
switchport access vlan 90
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 70
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 80
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 90
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
end

```

Switch-C-2:

```

!
hostname Switch-C-2
!
ip arp inspection vlan 70,80,90
ip arp inspection validate src-mac dst-mac ip
!
ip dhcp snooping vlan 70,80,90
no ip dhcp snooping information option
ip dhcp snooping
!
spanning-tree mode pvst

```

```
spanning-tree extend system-id
spanning-tree vlan 80 priority 24576
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90
ip arp inspection trust
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk
!
interface FastEthernet0/3
switchport access vlan 70
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/4
switchport access vlan 80
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/5
switchport access vlan 90
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/6
switchport access vlan 70
switchport mode access
!
interface FastEthernet0/7
switchport access vlan 80
switchport mode access
!
interface FastEthernet0/8
switchport access vlan 90
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 70
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 80
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 90
switchport mode access
!
interface Vlan1
no ip address
shutdown
```

```
!  
line con 0  
!  
line vty 0 4  
login  
line vty 5 15  
login  
!  
end
```

Switch-C-3:

```
!  
hostname Switch-C-3  
!  
ip arp inspection vlan 70,80,90  
ip arp inspection validate src-mac dst-mac ip  
!  
ip dhcp snooping vlan 70,80,90  
no ip dhcp snooping information option  
ip dhcp snooping  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
spanning-tree vlan 70,80 priority 20480  
spanning-tree vlan 90 priority 24576  
!  
interface FastEthernet0/1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90  
ip arp inspection trust  
switchport mode trunk  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90  
switchport mode trunk  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90  
ip arp inspection trust  
ip dhcp snooping trust  
switchport mode trunk  
!  
interface Vlan1  
no ip address  
shutdown  
!  
line con 0  
!  
line vty 0 4  
login  
line vty 5 15  
login  
!
```

end

Switch-D-1:

```
!  
hostname Switch-D-1  
!  
ip arp inspection vlan 100,110,120  
ip arp inspection validate src-mac dst-mac ip  
!  
ip dhcp snooping vlan 100,110,120  
no ip dhcp snooping information option  
ip dhcp snooping  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface FastEthernet0/1  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 100,110,120  
  ip arp inspection trust  
  switchport mode trunk  
!  
interface FastEthernet0/2  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 100,110,120  
  ip arp inspection trust  
  ip dhcp snooping trust  
  switchport mode trunk  
!  
interface FastEthernet0/3  
  switchport access vlan 100  
  switchport mode access  
  switchport port-security  
  switchport port-security aging time 60  
!  
interface FastEthernet0/4  
  switchport access vlan 110  
  switchport mode access  
  switchport port-security  
  switchport port-security aging time 60  
!  
interface FastEthernet0/5  
  switchport access vlan 120  
  switchport mode access  
  switchport port-security  
  switchport port-security aging time 60  
!  
interface FastEthernet0/6  
  switchport access vlan 100  
  switchport mode access  
!  
interface FastEthernet0/7  
  switchport access vlan 110  
  switchport mode access  
!  
interface FastEthernet0/8  
  switchport access vlan 120  
  switchport mode access
```

```

!
interface FastEthernet0/9
switchport access vlan 100
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 110
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 120
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
end

```

Switch-D-2:

```

!
hostname Switch-D-2
!
ip arp inspection vlan 100,110,120
ip arp inspection validate src-mac dst-mac ip
!
ip dhcp snooping vlan 100,110,120
no ip dhcp snooping information option
ip dhcp snooping
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120
ip arp inspection trust
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk
!
interface FastEthernet0/3
switchport access vlan 100
switchport mode access
switchport port-security
switchport port-security aging time 60

```

```

!
interface FastEthernet0/4
switchport access vlan 110
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/5
switchport access vlan 120
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/6
switchport access vlan 100
switchport mode access
!
interface FastEthernet0/7
switchport access vlan 110
switchport mode access
!
interface FastEthernet0/8
switchport access vlan 120
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 100
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 110
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 120
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
end

```

Switch-D-3:

```

!
hostname Switch-D-3
!
ip arp inspection vlan 100,110,120
ip arp inspection validate src-mac dst-mac ip
!
ip dhcp snooping vlan 100,110,120

```

```

no ip dhcp snooping information option
ip dhcp snooping
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120
ip arp inspection trust
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120
ip arp inspection trust
switchport mode trunk
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
end

```

Router-A:

```

!
hostname Router-A
!
ip dhcp excluded-address 192.168.1.30
ip dhcp excluded-address 192.168.1.62
ip dhcp excluded-address 192.168.1.94
!
ip dhcp pool VLAN10
network 192.168.1.0 255.255.255.224
default-router 192.168.1.30
ip dhcp pool VLAN20
network 192.168.1.32 255.255.255.224
default-router 192.168.1.62
ip dhcp pool VLAN30
network 192.168.1.64 255.255.255.224
default-router 192.168.1.94
!
ip cef
no ipv6 cef

```

```
!  
license udi pid CISCO2911/K9 sn FTX15240R36-  
!  
spanning-tree mode pvst  
!  
interface GigabitEthernet0/0  
no ip address  
ip nat inside  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/0.10  
encapsulation dot1Q 10  
ip address 192.168.1.30 255.255.255.224  
ip access-group To_Network_C in  
ip nat inside  
!  
interface GigabitEthernet0/0.20  
encapsulation dot1Q 20  
ip address 192.168.1.62 255.255.255.224  
ip access-group To_Network_C in  
ip nat inside  
!  
interface GigabitEthernet0/0.30  
encapsulation dot1Q 30  
ip address 192.168.1.94 255.255.255.224  
ip access-group To_Network_C in  
ip nat inside  
!  
interface GigabitEthernet0/1  
ip address 203.0.113.1 255.255.255.252  
ip nat outside  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
ip address 203.0.113.14 255.255.255.252  
ip nat outside  
duplex auto  
speed auto  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 203.0.113.0 0.0.0.3 area 0  
network 192.168.1.0 0.0.0.31 area 0  
network 192.168.1.32 0.0.0.31 area 0  
network 192.168.1.64 0.0.0.31 area 0  
network 203.0.113.12 0.0.0.3 area 0  
!  
ip nat inside source list 1 interface GigabitEthernet0/1 overload  
ip nat inside source list 2 interface GigabitEthernet0/2 overload  
ip classless  
!  
ip flow-export version 9  
!  
  
access-list 1 permit 192.168.1.0 0.0.0.31
```

```

access-list 1 permit 192.168.1.32 0.0.0.31
access-list 1 permit 192.168.1.64 0.0.0.31
access-list 1 deny any
ip access-list extended To_Network_C
deny ip 192.168.1.0 0.0.0.127 192.168.1.192 0.0.0.63
deny ip 192.168.1.0 0.0.0.127 192.168.2.0 0.0.0.31
permit ip any any
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
end

```

Router-B:

```

!
hostname Router-B
!
ip dhcp excluded-address 192.168.1.126
ip dhcp excluded-address 192.168.1.158
ip dhcp excluded-address 192.168.1.190
!
ip dhcp pool VLAN40
network 192.168.1.96 255.255.255.224
default-router 192.168.1.126
ip dhcp pool VLAN50
network 192.168.1.128 255.255.255.224
default-router 192.168.1.158
ip dhcp pool VLAN60
network 192.168.1.160 255.255.255.224
default-router 192.168.1.190
!
ip cef
no ipv6 cef
!
license udi pid CISCO2911/K9 sn FTX15247R9B-
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0
no ip address
ip nat inside
duplex auto
speed auto
!
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 192.168.1.126 255.255.255.224
!
interface GigabitEthernet0/0.50

```

```

encapsulation dot1Q 50
ip address 192.168.1.158 255.255.255.224
!
interface GigabitEthernet0/0.60
encapsulation dot1Q 60
ip address 192.168.1.190 255.255.255.224
!
interface GigabitEthernet0/1
ip address 203.0.113.2 255.255.255.252
ip nat outside
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 203.0.113.5 255.255.255.252
ip nat outside
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 203.0.113.0 0.0.0.3 area 0
network 192.168.1.96 0.0.0.31 area 0
network 192.168.1.128 0.0.0.31 area 0
network 192.168.1.160 0.0.0.31 area 0
network 203.0.113.4 0.0.0.3 area 0
!
ip nat inside source list 1 interface GigabitEthernet0/1 overload
ip nat inside source list 2 interface GigabitEthernet0/2 overload
ip classless
!
ip flow-export version 9
!
access-list 1 permit 192.168.1.0 0.0.0.31
access-list 1 permit 192.168.1.32 0.0.0.31
access-list 1 permit 192.168.1.64 0.0.0.31
access-list 2 permit 192.168.1.0 0.0.0.31
access-list 2 permit 192.168.1.32 0.0.0.31
access-list 2 permit 192.168.1.64 0.0.0.31
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
end

```

Router-C:

```

!
hostname Router-C
!
ip dhcp excluded-address 192.168.1.222

```

```
ip dhcp excluded-address 192.168.1.254
ip dhcp excluded-address 192.168.2.30
!
ip dhcp pool VLAN70
network 192.168.1.192 255.255.255.224
default-router 192.168.1.222
ip dhcp pool VLAN80
network 192.168.1.224 255.255.255.224
default-router 192.168.1.254
ip dhcp pool VLAN90
network 192.168.2.0 255.255.255.224
default-router 192.168.1.30
!
ip cef
no ipv6 cef
!
license udi pid CISCO2911/K9 sn FTX15245210-
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.70
encapsulation dot1Q 70
ip address 192.168.1.222 255.255.255.224
ip access-group To_Network_A in
ip nat inside
!
interface GigabitEthernet0/0.80
encapsulation dot1Q 80
ip address 192.168.1.254 255.255.255.224
ip access-group To_Network_A in
ip nat inside
!
interface GigabitEthernet0/0.90
encapsulation dot1Q 90
ip address 192.168.2.30 255.255.255.224
ip access-group To_Network_A in
ip nat inside
!
interface GigabitEthernet0/1
ip address 203.0.113.6 255.255.255.252
ip nat outside
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 203.0.113.9 255.255.255.252
ip nat outside
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
```

```

network 203.0.113.4 0.0.0.3 area 0
network 192.168.1.192 0.0.0.31 area 0
network 192.168.1.224 0.0.0.31 area 0
network 192.168.2.0 0.0.0.31 area 0
network 203.0.113.8 0.0.0.3 area 0
!
ip nat inside source list 1 interface GigabitEthernet0/1 overload
ip nat inside source list 2 interface GigabitEthernet0/2 overload
ip classless
!
ip flow-export version 9
!
access-list 1 permit 192.168.1.192 0.0.0.31
access-list 1 permit 192.168.1.224 0.0.0.31
access-list 1 permit 192.168.2.0 0.0.0.31
access-list 1 deny any
access-list 2 permit 192.168.1.192 0.0.0.31
access-list 2 permit 192.168.1.224 0.0.0.31
access-list 2 permit 192.168.2.0 0.0.0.31
access-list 2 deny any
ip access-list extended To_Network_A
deny ip 192.168.1.192 0.0.0.63 192.168.1.0 0.0.0.31
deny ip 192.168.1.192 0.0.0.63 192.168.1.32 0.0.0.31
deny ip 192.168.1.192 0.0.0.63 192.168.1.64 0.0.0.31
deny ip 192.168.2.0 0.0.0.31 192.168.1.0 0.0.0.31
deny ip 192.168.2.0 0.0.0.31 192.168.1.32 0.0.0.31
deny ip 192.168.2.0 0.0.0.31 192.168.1.64 0.0.0.31
permit ip any any
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
end

```

Router-D:

```

!
hostname Router-D
!
ip dhcp excluded-address 192.168.2.62
ip dhcp excluded-address 192.168.2.94
ip dhcp excluded-address 192.168.2.126
!
ip dhcp pool VLAN100
network 192.168.2.32 255.255.255.224
default-router 192.168.2.62
ip dhcp pool VLAN110
network 192.168.2.64 255.255.255.224
default-router 192.168.2.94
ip dhcp pool VLAN120
network 192.168.2.96 255.255.255.224
default-router 192.168.2.126
!
ip cef
no ipv6 cef

```

```
!  
license udi pid CISCO2911/K9 sn FTX1524JO4Q  
!  
spanning-tree mode pvst  
!  
interface GigabitEthernet0/0  
no ip address  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/0.100  
encapsulation dot1Q 100  
ip address 192.168.2.62 255.255.255.224  
!  
interface GigabitEthernet0/0.110  
encapsulation dot1Q 110  
ip address 192.168.2.94 255.255.255.224  
!  
interface GigabitEthernet0/0.120  
encapsulation dot1Q 120  
ip address 192.168.2.126 255.255.255.224  
!  
interface GigabitEthernet0/1  
ip address 203.0.113.10 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
ip address 203.0.113.13 255.255.255.252  
duplex auto  
speed auto  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 203.0.113.8 0.0.0.3 area 0  
network 203.0.113.12 0.0.0.3 area 0  
network 192.168.2.32 0.0.0.31 area 0  
network 192.168.2.64 0.0.0.31 area 0  
network 192.168.2.96 0.0.0.31 area 0  
!  
ip classless  
!  
ip flow-export version 9  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
login  
!  
end
```